

audience that is quickly growing. This might not seem very new. There have been mobile websites for a long time you might say. Ah! But the fact is there aren't two different webs for mobile and desktop. They aren't two separate worlds any more! You had mobiles and desktops, then you got netbooks, now we also have TVs and tablets, what next? Can you create a separate website for each? What about the iPad 3 and iPhone 4, which have small screens but higher resolution?

Responsive design is important because it forces you to think about these things. A well designed responsive website is already ready for that new device coming out tomorrow with that weird aspect ratio and unconventional resolution.

How to go responsive

While there is a continuum of devices and resolutions going from 3-inch to 60-inch with nearly every step in between, it is obviously not possible to respond to every pixel change. Responsive design is thus fluid, not based on a fixed width, but rather the width of the browser, such that as the size of the browser changes, the elements of the page scale with it.

However, as we have said before, this isn't enough. Just scaling down the page will make it difficult to navigate or read. So while resizing is fine within reasonable limits, after a point we should consider adapting the layout to the new size.

You can look at the standard size of a tablet device, and change the layout of your website to a tablet-friendly layout when the size of the browser falls below that limit. This can be done purely in CSS using a feature called media queries. CSS media queries are a powerful way to take the same HTML structure and style it differently based on the form factor of the device being used to view the page.

Here pixel density is also an important consideration. The iPad 3, for example, has a display resolution that is higher than most desktops, but sending it the desktop version of your site is a bad idea, since it is a much smaller screen than a desktop of a similar resolution (9.7-inch compared to 27-inch). Luckily, CSS can query this as well.

Now when the screen size moves to a smaller size still you know that you have reached mobile territory, and you need to adapt your layout once again. At this point, you are best off with a single-column layout. Once again the screen width isn't enough, because a device like the iPhone 4 has a display resolution that is close to tablet territory. Similarly you can use CSS media queries to detect when the website is being viewed on a TV, and change the layout and style accordingly. You need not restrict yourself to predefined form factors of mobile,

*pointers

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*The Web platform's cutting edge

>> Dimitri Glazkov, and Alex Komoroske share what they do, and how you can start playing with these powerful technologies to build robust, reusable, and packageable components.

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*shortcomings

>> These aren't shortcomings of the Responsive Design "philosophy," but rather the shortcomings of current web standards and technologies that make creating a good responsive website difficult. One of the biggest sore spots when it comes to responsive design, is the matter of responsive images. Let us warn straight away that this problem has not been completely solved yet, although there are some clever solutions out there.

While it is an easy matter to deal with image assets used by a website's design, such as the headers, logos and other elements of design. One simply has to use the correct image depending on the media query. However how do you deal with images embedded in HTML using the tag? Turns out this is not as simple as one might think.

If you use JavaScript to insert an image, none of our images will work without JavaScript. If you have JavaScript change the image source based on the the device size, it will fetch both the smaller and larger version of the image.

This can be a long topic in itself but we'll spare you the details and just tell you that there is no solution that works entirely in the browser, across all browsers. Current solutions rely on a combination of client-side JavaScript code, and server-side code to select and send the best asset for the device requesting the image.

Worry not, there is ongoing work to create standards that deal with this issue, since it is likely to be central to the web in the future.

Another big shortcoming of current standards is that while CSS can be responsive to screen size and pixel density using media queries, it is not possible to take into account the power of the device, or the bandwidth of the connection. For now the common assumption is that the smaller the screen, the slower the device and its network connection. This isn't always true. Take for example the latest third-generation iPad, it has a higher resolution than most desktops, but is less powerful and likely has a slower internet connection. Similarly even the most powerful of desktops in India are often have rather poor internet connectivity.

tablet, desktop and TV; you can add as many breakpoints as you wish and adjust based on those criteria.

As you can see, Responsive Design isn't just based on the use of any particular technology; but, rather at using everything at your disposal to ensure that your website is adaptable to any kind of device. >

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